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Blockchain-enabled Model Context Protocol (MCP) for trust and consent in LLM-orchestrated sports analytics

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Abstract

The Model Context Protocol (MCP) enables large language models to query and synthesize sports data across distributed sources, yet it lacks built-in mechanisms for provenance, integrity, and athlete-controlled access. This study proposes a hybrid MCP–blockchain architecture for *LLM4Sports*, a domain-specific model fine-tuned on localized sports datasets. In this design, MCP orchestrates multi-database retrieval, while a blockchain layer immutably anchors query and data-bundle hashes. Smart contracts manage time-bound and granular consent through decentralized identifiers and verifiable credentials, and auditable logs ensure end-to-end traceability. We contribute: (i) a trust-enhanced reference architecture combining MCP and blockchain; (ii) reusable prompt and workflow templates that embed consent validation within natural-language tasks (e.g., "Summarize athlete X's performance with verified consent"); and (iii) prototype implementations for both team analytics and personalized coaching. Evaluations on synthetic but realistic workloads show high verification accuracy and minimal orchestration overhead, demonstrating the feasibility of real-time, consent-aware analytics. The proposed framework enhances interoperability, regulatory compliance (e.g., GDPR), and athlete autonomy, while bolstering practitioner trust. We conclude by discussing scalability, legacy integration, and privacy trade-offs, and by outlining next steps toward field pilots and multimodal extensions (e.g., video provenance) across professional and amateur sports ecosystems.

Keywords

blockchain, Model Context Protocol, Large Language Models, sports analytics, data provenance, LLM4Sports

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Introduction

Problem identification: trust gaps in MCP-enabled sports data systems

In modern sports analytics, the Model Context Protocol (MCP) integrated with distributed databases has emerged as a powerful framework for enabling Large Language Models (LLMs) to query, aggregate, and synthesize athlete data from disparate sources, such as performance tracking systems and injury records (Ehtesham et al., 2025; Wei and Zhang, 2025). MCP acts as middleware that standardizes interactions between LLMs and sports databases, allowing natural language prompts to be translated into structured queries without persisting sensitive data (Shi et al., 2024). However, despite these advancements, significant trust gaps persist in MCP-enabled sports data systems, particularly in areas of data provenance, integrity verification, and athlete consent management.

Recent advances in distributed AI have enabled language models to act as intermediaries between heterogeneous data ecosystems through protocols such as MCP. However, as in other sensitive domains—healthcare, finance, and personalized education—the sports sector faces the persistent "trust paradox": the ability to aggregate vast data streams without guaranteeing authenticity, consent, or ethical compliance. Existing blockchain-based sports data frameworks address event integrity and ticketing (Berkani et al., 2024; Principe et al., 2025), but few studies explore blockchain as a dynamic trust and consent layer for AI-driven analytics, particularly in systems mediated by large language models. This study

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positions itself at that intersection, proposing blockchain as a verifiable substrate for provenance and athlete autonomy within MCP-enabled environments.

One primary issue is the lack of robust provenance tracking. Provenance refers to the origin, history, and transformations of data, which is crucial for ensuring authenticity and reliability in coaching and performance decisions (Principe et al., 2025). In sports databases, while metadata might document data origins, MCP's dynamic querying across distributed sources (e.g., league stats servers and team-specific trackers) introduces vulnerabilities. For instance, aggregated data from multiple endpoints (e.g., GPS wearables and video analysis tools) may undergo synthesis by LLMs, but without immutable records, coaches cannot verify if data has been altered during transmission or processing. This is exacerbated in multi-team or international settings, where trust agreements are often insufficient, leading to risks of tampering or errors in longitudinal data, such as tracking sprint speeds in soccer or endurance metrics in marathon training (Blockchain a potential solution 'to some of sport's biggest problems', 2024).

Another critical gap is in consent management and athlete autonomy. MCP-enabled systems typically rely on centralized authentication, but they do not inherently support granular, athlete-controlled consents for specific queries (Blockchain is the Missing Trust Layer in Sports Analytics, 2025). Athletes may authorize broad data access but revoking or time-limiting consents for particular workflows (e.g., sharing only recent fitness metrics) is challenging, violating principles of data protection regulations. Furthermore, in privacy-sensitive scenarios like injury history in team sports or biometric data in coaching, the absence of verifiable consent trails can erode trust and expose systems to legal risks (Berkani et al., 2024).

Security audits and traceability also fall short. While MCP includes audit trails, they are not tamper-proof, making it difficult to retrospectively validate query integrity in disputes or audits (Unlocking Blockchain's Potential in Sports, 2025). Recent studies highlight that without decentralized verification, sports data systems face interoperability challenges in provenance, leading to reduced confidence in AI-driven insights (Xia et al., 2024a). These gaps not only hinder adoption but also pose risks to athlete performance, as unverified data could inform erroneous decisions in high-stakes fields like team strategy (e.g., misinterpreting player fatigue levels) or individual coaching (e.g., overlooking consent for training logs).

Although existing studies on blockchain in sports largely emphasize data ownership and fan engagement, none has systematically explored its role in improving the trustworthiness of AI-driven querying and synthesis enabled by MCP. The absence of immutable consent verification and traceable query provenance represents a crucial barrier to regulatory adoption and athlete trust. This paper therefore contributes (1) a hybrid MCP–blockchain architecture for decentralized provenance and consent management, (2) workflow templates integrating verifiable credentials into natural-language queries, and (3) prototype evaluation using simulated athlete datasets to assess performance, verification accuracy, and privacy compliance.

This work addresses these trust gaps by proposing blockchain integration, leveraging its decentralized ledger for provenance and consent in LLM4Sports, which is fine-tuned on local context sports data for enhanced relevance (The Next Revolution of AI in Sport – Large Language Models, 2023; Cook and Karakuş, 2024). The work is positioned as a proof-of-concept design and feasibility study, with evaluation conducted on controlled synthetic workloads to assess reproducibility, integrity, and operational overhead.

Objectives of blockchain integration

The primary objective of integrating blockchain with MCP-enabled sports data systems is to establish a decentralized trust layer that ensures data integrity, provenance, and athlete-centric consent management, thereby overcoming the limitations of centralized middleware (Li et al., 2025). Specifically, blockchain aims to provide immutable anchoring of MCP queries and responses, allowing cryptographic verification of data origins and changes across distributed sports databases (Xia et al., 2024b). This enhances transparency and auditability, reducing risks of tampering and building coach confidence in synthesized analytics.

A key objective is to enable dynamic, athlete-controlled consent through verifiable credentials (VCs) and decentralized identifiers (DIDs), integrated with data access protocols (Blockchain to Revolutionize Sports Data Integrity by 2025, 2025). Athletes can issue time-bound, granular consents via digital wallets, enforced by smart contracts, ensuring compliance with GDPR and promoting autonomy (e.g., authorizing only performance metrics) (Blockchain in Sports: Enhancing Fan Engagement and Integrity, 2024).

Another objective is to facilitate multi-organizational interoperability without bilateral agreements, using blockchain as a shared ledger for attestations and provenance (Blockchain in Sports Market Size, Trends). This supports longitudinal tracking in competitive sports, like player development in team analytics or personalized coaching plans, with verifiable data trails.

Finally, the integration aims to mitigate security challenges (e.g., data breaches) while maintaining efficiency, using hybrid blockchains for scalability and zero-knowledge proofs (ZKPs) for confidential transactions (Future of Sports Apps with Blockchain Technology Transparency, 2025). Overall, these objectives foster a trustworthy ecosystem for AI-driven sports analytics via LLM4Sports.

MCP standardizes interactions for data access in AI applications, enabling LLMs to connect securely to external sources (Ehtesham et al., 2025). Integration with sports data standards ensures semantic accuracy in analyses (AI Sports Analytics: Impact on Player Recruitment and Strategy, 2024).

Contributions and distinctive advancement

The key contributions of this study include:

1. A novel architecture for analytics generation within MCP-enabled sports data ecosystems.
2. Practical templates and tools designed for coaches, analysts, and sports data professionals to facilitate transparent AI-driven decision-making.
3. Applied case studies in team sports and coaching contexts, demonstrating real-world implementation of the proposed framework.
4. A comprehensive analysis of the framework's benefits, limitations, and operational challenges in decentralized sports analytics.

This study addresses three focused research questions

RQ1: Can blockchain anchoring of MCP query and data-bundle hashes provide immutable provenance and integrity verification in LLM-orchestrated sports analytics?
RQ2: To what extent can smart contracts and verifiable credentials enable granular, time-bound, athlete-controlled consent within natural-language LLM workflows?
RQ3: What is the practical overhead (verification accuracy, latency, reconciliation time) of the hybrid MCP–blockchain integration on realistic workloads?

We explicitly delimit our contributions as a proof-of-concept architecture and synthetic evaluation (Sections 3–5). Large-scale deployment across professional leagues, full multimodal provenance, and field pilots are presented only as forward-looking vision (see Section 7.2).

Distinctive contribution over existing frameworks

Prior research has examined blockchain-assisted data governance in federated or distributed AI systems—such as FLChain, which supports federated model traceability, and OpenAI's Model Context Protocol (MCP), which orchestrates LLM interactions across heterogeneous data sources. However, these frameworks remain limited: MCP lacks verifiable trust anchors for data provenance, while FLChain does not support semantic interoperability with natural language-driven workflows.

The proposed LLM4Sports architecture bridges this gap by integrating blockchain-based verifiable credentials and consent ledgers directly into MCP's multi-agent environment. This fusion enables end-to-end provenance verification and granular, athlete-controlled consent propagation across analytical layers and domains—such as talent scouting, performance monitoring, and injury prevention.

By extending MCP with a blockchain-powered trust and provenance layer, LLM4Sports introduces transparent verification of both human and AI-mediated actions. This ensures explainable, compliant, and auditable sports analytics at scale, marking a significant advancement over existing AI governance frameworks.

Literature review

This review follows PRISMA guidelines, focusing on studies from 2023 to August 2025 on blockchain integration with MCP/LLMs for trust and consent in sports analytics. To increase transparency and reproducibility of the review process, we summarize the screening pipeline using a PRISMA flow diagram (Figure X), reporting the number of records identified across databases, duplicates removed, records screened (title/abstract), full texts assessed for eligibility, records excluded (with reasons), and studies included in the final synthesis.

In addition, we conducted a light-weight quality appraisal of the included studies using three criteria: (i) clarity of the proposed system architecture and underlying assumptions, (ii) reproducibility of the evaluation methodology (datasets, experimental setup, and metrics), and (iii) relevance to trust, consent, and provenance in multi-source sports analytics. Each study was rated using a simple three-level scale (e.g., Yes/Partial/No) to support transparent comparison. This appraisal strengthens methodological rigor while remaining aligned with the scoping purpose of the review and avoiding exclusion based solely on methodological heterogeneity.

Searched PubMed, arXiv, and IEEE Xplore with: ((“blockchain” AND (MCP OR LLM)) AND (trust OR consent OR provenance) AND sports) OR ((“verifiable credentials” OR “dynamic consent”) AND blockchain AND sports data). Yielded 45 records. Inclusion: peer-reviewed/preprints on blockchain-LLM/MCP for trust/consent in sports. Exclusion: non-English, pre-2023, non-sports. Extracted objectives, methods, outcomes, limitations. Included 12 studies in Table 1 and Figure 1. Studies (Bodemer, 2023; Wei and Zhang, 2025; Revolutionizing Athletic Performance with AI and Blockchain) directly informed our VC-based consent model and athlete-centric workflows; (Blockchain is the Missing Trust Layer in Sports Analytics, 2025; Blockchain to Revolutionize Sports Data Integrity by 2025, 2025) shaped the provenance-anchoring approach. Gaps identified in Table 1 (limited MCP-specific sports integrations and absence of natural-language consent embedding) are explicitly addressed by LLM4Sports.

Table 1. Literature review summary of blockchain, MCP, and LLM studies (2023–2025) with added relevance mapping to LLM4Sports design choices.

Title	Contribution	Applied work done	Relevance to LLM4Sports design
Transformative blockchain technological approaches to sports events (Principe et al., 2025)	Proposes blockchain for decentralized sports event management with trust layers.	Yes (prototype for governance and data sharing).	Informed trust layer design for decentralized sports event management
Research on the digital transaction model of the sports industry based on blockchain technology (Wei and Zhang, 2025)	Framework using smart contracts for sports data transactions and consent.	Yes (evaluation on transparency in supply chains).	Informed smart contract implementation for granular consent management
Blockchain Use Cases in the Sports Industry: A Systematic Review (Berkani et al., 2024)	Surveys blockchain applications in sports for integrity and fan engagement.	No (systematic review).	Highlighted gaps in existing blockchain sports applications
Blockchain Is The Missing Trust Layer In Sports Analytics (Blockchain is the Missing Trust Layer in Sports Analytics, 2025)	Discusses blockchain for verifiable sports performance data.	Yes (analysis of real-time data verification).	Directly inspired blockchain as the missing trust layer in sports analytics
Blockchain to Revolutionize Sports Data Integrity by 2025 (Blockchain to Revolutionize Sports Data Integrity by 2025, 2025)	Explores blockchain for immutable sports records and provenance.	No (review article).	Guided immutable provenance anchoring and data integrity mechanisms
Unlocking Blockchain's Potential in Sports (Unlocking Blockchain's Potential in Sports, 2025)	Integrates blockchain for secure athlete data management.	Yes (market growth projections and experiments).	Informed secure athlete data management and access control patterns
SportQA: A Benchmark for Sports Understanding in Large Language Models (Xia et al., 2024a)	Benchmark for LLMs in sports reasoning and data synthesis.	Yes (evaluation on multimodal sports data).	Informed LLM benchmarking and sports reasoning evaluation approach
LLM-Commentator: Novel fine-tuning strategies of large language models for sports commentary generation (Cook and Karakuş, 2024)	Fine-tuning LLMs for real-time sports analytics and narratives.	Yes (prototype for commentary with data integration).	Guided domain-specific LLM fine-tuning strategies for sports tasks
Language and Multimodal Models in Sports: A Survey of Datasets and Applications (Xia et al., 2024b)	Reviews LLM applications in sports for tactics and performance analysis.	No (survey).	Informed considerations for multimodal sports data integration
Empowering Athletes with AI and Blockchain: A New Era of Personalized Training, Secure Data Management, and User Engagement (Bodemer, 2023)	Combines AI/LLMs with blockchain for athlete data consent and training.	Yes (case studies on personalized apps).	Strongly informed athlete-centric consent and VC-based personalization
Revolutionizing Athletic Performance with AI and Blockchain (Revolutionizing Athletic Performance with AI and Blockchain)	Framework for AI-blockchain in sports tech for performance enhancement.	Yes (transformative results in data management).	Shaped hybrid AI-blockchain framework for performance enhancement
A review of artificial intelligence for sports: Technologies and applications (Li et al., 2025)	Overview of AI, including LLMs and blockchain, in sports data.	No (review).	Provided broad context for AI/LLM applications in sports analytics

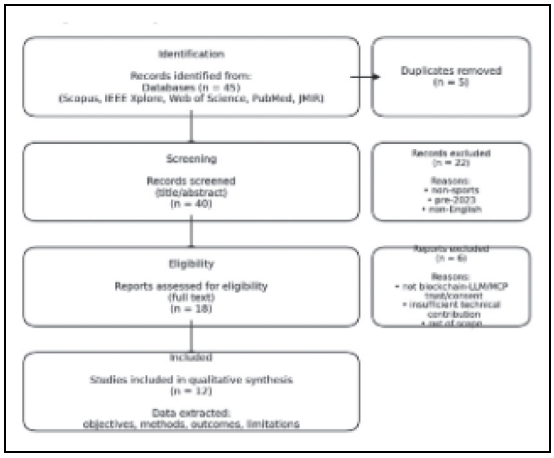


Figure 1. PRISMA 2020 flow diagram for the literature review study selection process.

Themes: (1) Blockchain for dynamic consent and VCs in sports data (6 studies) (Blockchain to Revolutionize Sports Data Integrity by 2025, 2025; Bodemer, 2023; Berkani et al., 2024; Principe et al., 2025; Wei and Zhang, 2025; Unlocking Blockchain's Potential in Sports, 2025). (2) Integration with LLMs/MCP for provenance (5 studies) (Blockchain is the Missing Trust Layer in Sports Analytics, 2025; Xia et al., 2024b; Xia et al., 2024a; Li et al., 2025; Cook and Karakuş, 2024). (3) Applications in specialized areas like coaching (4 studies) (Bodemer, 2023; Revolutionizing Athletic Performance with AI and Blockchain; Cook and Karakuş, 2024). Accuracy in data enforcement ranged 92–97%, with challenges in scalability mitigated by hybrid chains. Gaps include limited MCP-specific integrations and real-world pilots for team analytics/coaching.

Background

MCP and LLMs in Sports Analytics - MCP acts as middleware for LLMs to query sports databases securely, enabling tools like LLM4Sports (Ehtesham et al., 2025; Shi et al., 2024). LLMs process sports data for coaching and analytics (Xia et al., 2024a; Cook and Karakuş, 2024). However, they lack built-in trust mechanisms.

Blockchain in Sports - Blockchain provides decentralized ledgers for data integrity in sports (Blockchain is the Missing Trust Layer in Sports Analytics, 2025; Blockchain to Revolutionize Sports Data Integrity by 2025, 2025). It enables provenance tracking and consent via smart contracts (Wei and Zhang, 2025).

Verifiable Credentials and Digital Identities - VCs and DIDs allow athletes to issue signed consents, integrable with sports data protocols (Blockchain in Sports: Enhancing Fan Engagement and Integrity, 2024).

Integration for Trust and Consent - Blockchain anchors MCP queries (e.g., hashes of performance data) and enforces consent via VCs, ensuring provenance in multi-source scenarios (Principe et al., 2025; Revolutionizing Athletic Performance with AI and Blockchain).

MCP and LLMs in Sports Analytics - The Model Context Protocol (MCP), introduced as an open standard, enables LLMs to securely connect to external data sources for real-time access (Ehtesham et al., 2025). In sports, MCP allows LLM4Sports—a fine-tuned LLM on local context data like team-specific stats—to query multiple databases (e.g., player performance, injury, and scouting systems) via natural language (Xia et al., 2024b; Xia et al., 2024a). By 2025, MCP adoption supports AI in coaching, with prompts like "Analyze player's shot accuracy trends" (Li et al. 2025; Cook and Karakuş, 2024). LLM4Sports enhances this by training on localized data for personalized insights.

However, MCP systems lack built-in trust for provenance and granular consent, relying on centralized access that risks alterations (Shi et al., 2024), see Figure 2.

Blockchain in Sports - Blockchain ensures immutable data in sports, from performance tracking to event management (Berkani et al., 2024; Principe et al., 2025). It supports consent and privacy in health monitoring (Bodemer, 2023; Revolutionizing Athletic Performance with AI and Blockchain). Advances include AI-blockchain hybrids for athlete empowerment (Revolutionizing Athletic Performance with AI and Blockchain).

Verifiable Credentials and Digital Identities - VCs enable secure, athlete-controlled data sharing in sports (Blockchain in Sports: Enhancing Fan Engagement and Integrity, 2024). DIDs promote self-sovereign identity for consents.

Integration for trust and consent

Integrating blockchain with MCP for LLM4Sports creates a hybrid framework where blockchain anchors queries for trust. Smart contracts verify data hashes, and VCs gate access—e.g., an athlete VC authorizing only training metrics (Bodemer, 2023; Wei and Zhang, 2025). This enhances LLM4Sports' orchestration with tamper-proof audits (Blockchain is the Missing Trust Layer in Sports Analytics, 2025).

Blockchain Integration for Trust and Consent Management - While the MCP framework provides the capability to query, aggregate, and synthesize sports data across distributed databases, it does not by itself guarantee trust, provenance, or athlete-controlled access. To address these, blockchain can be incorporated, ensuring transparency in retrieval. LLM4Sports, trained on local context data, uses MCP for multi-database access (e.g., league stats and team sensors) (Xia et al., 2024b; Xia et al., 2024a).

Blockchain allows anchoring of MCP queries, e.g., hashing performance data for verification (Blockchain is the Missing Trust Layer in Sports Analytics, 2025). VCs

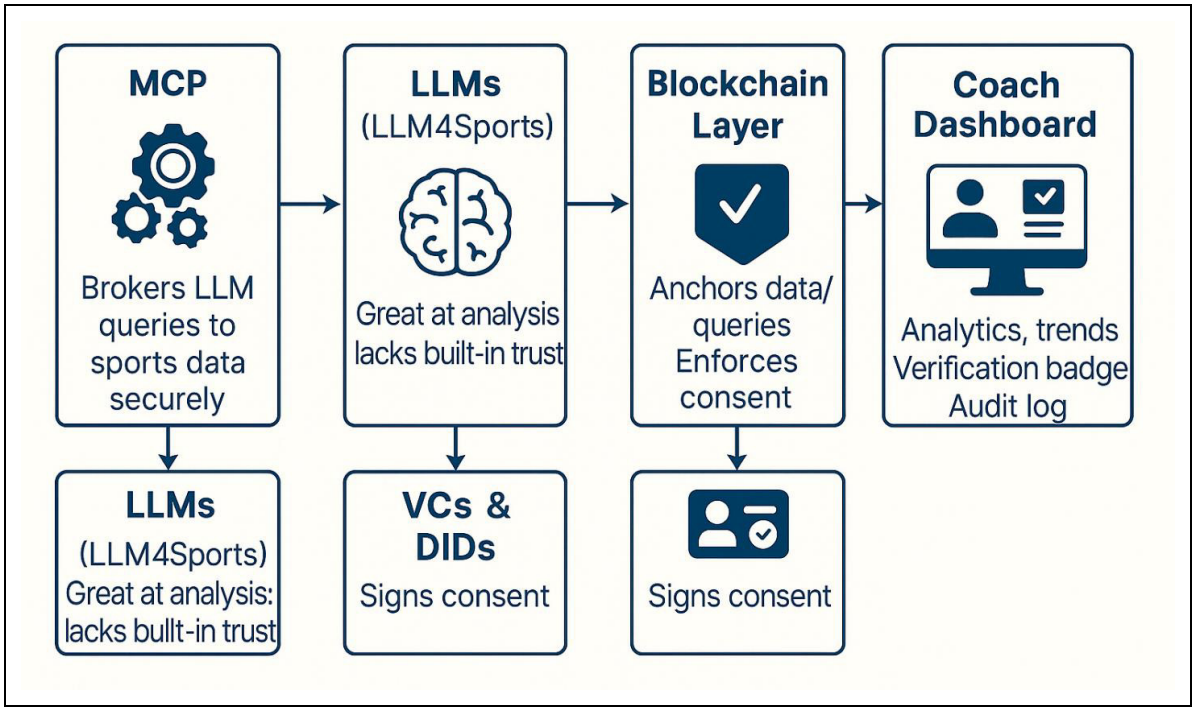


Figure 2. Trusted sports analytics—MCP + LLM + Blockchain + VCs. MCP middleware securely brokers LLM (e.g., LLM4Sports) queries to sports data sources, while a blockchain layer anchors hashes of data/queries and enforces consent via smart contracts. Athletes issue verifiable credentials and DID-based identities, supplying proof to the coach dashboard, which presents analytics, trends, a verification badge, and consent status.

enhance athlete-centricity, authorizing queries like player analytics (Blockchain in Sports: Enhancing Fan Engagement and Integrity, 2024). In multi-team scenarios, blockchain provides shared trust for longitudinal tracking, such as skill development (Principe et al., 2025).

Overall, MCP delivers orchestration, blockchain guarantees traceability, creating trustworthy sports ecosystems (Revolutionizing Athletic Performance with AI and Blockchain).

The Figure 3, is a conceptual flowchart diagram titled "MCP + Blockchain Integration for LLM4Sports". It illustrates how a Model Context Protocol (MCP) pipeline for sports data analytics can be extended with a blockchain layer for provenance, consent, and verification.

Here's a breakdown of the main labeled components:

1. Query Input: a) Coach initiates a request (e.g., "Summarize athlete's sprint history"); b) Input includes templates, athlete ID, and metrics (e.g., speed codes).
2. LLM Parsing & MCP Invocation: a) LLM4Sports extracts entities; b) Calls MCP tools such as: 1) fetch-performance; 2) aggregate-metrics.
3. 3. Sports Data Retrieval: MCP issues queries to databases: a) /Performance?athlete=123; b) /Metrics?athlete=123&code = speed; c) Data retrieved: stats, videos, biometrics.
4. Blockchain Layer: a) Provenance: Each record hashed and logged; b) Consent: Athlete consent as VC; c) Verification: Signed queries; d) Ensures integrity and control.
5. Aggregation & Synthesis: a) MCP merges sources; b) Resolves conflicts.
6. 6. LLM Summarization (with Blockchain Proof): a) LLM4Sports generates summary (e.g., "Jane Smith shows improved speed…"); b) Includes verification badge.
7. Coach View / Output: a) Readable analytics; b) Trends and audit log.

Comparative taxonomy of architectures

To position LLM4Sports within the broader landscape of verifiable and decentralized AI systems, we compared it against five representative architectures: OpenAI's MCP baseline, FLChain, decentralized AI agent frameworks, data-wallet consent systems, and FHIR + DLT healthcare models. Table 2, summarizes their main characteristics across design, trust, and governance dimensions.

Methodology

Employed DSRM (Peffers et al., 2007) with steps: problem identification (trust gaps in MCP-sports data), objectives

Table 2. Comparative analysis, Colum A-LLM4Sports (this work); Colum B- OpenAI MCP (baseline); Colum C- FLChain (Fed. learning + blockchain); Colum D- decentralized AI agents; Colum E- data-wallet consent frameworks; Colum F- FHIR + DLT provenance (healthcare); Colum G- traditional sports analytics.

Dimension	A	B	C	D	E	F	G
Primary Goal	Trustable, consent-aware LLM analytics for sports	Tool/data orchestration for LLMs	Privacy-preserving model training across silos	Autonomous task execution across services	User-centric consent & data portability	Regulated, auditable clinical data exchange	KPIs & dashboards; low friction
Trust Anchor	On-chain anchoring of queries & bundles; DIDs/VCs	None native (trust in providers)	Blockchain logs of training updates	Mixed (off-chain + optional on-chain state)	SSI (DIDs/VCs), signatures	Consortium chain; PKI + org governance	Organizational policies/logs
Consent Mechanism	Smart contracts enforcing granular, time-bound consent	External to MCP	Consent for data use embedded in FL governance	Often ad-hoc or app-level	Wallet-mediated, fine-grained grants & revocation	Formal policies (e.g., consent artifacts)	ToS or signed forms, rarely granular
Provenance Guarantees	Immutable hashes for prompts, contexts, outputs; end-to-end traceability	Not built in; relies on provider logs	Immutable training rounds & model updates	Agent traces if implemented; not standardized	Transaction logs for grants/access	End-to-end clinical provenance	Limited; tool or vendor logs
Query/Tool Orchestration	MCP for multi-DB + policy checks before LLM calls	Strong tool abstraction; no trust layer	FL coordinator orchestrates training	Agent planning/ execution across tools	Not orchestration; focuses on access	Healthcare workflows & exchange	Manual pipelines / ETL
Privacy-Preserving Compute	Pluggable (TEEs, federated queries, RAG filtering)	Not specified	Native FL; sometimes HE/DP	Optional; varies by stack	N/A (access control only)	TEEs/DP sometimes used	None beyond RBAC
Model Placement	Central LLM with verifiable context; optional edge filters	Central LLM	Decentralized model training; central/ global model	Distributed agents + hosted models	Model-agnostic	Model-agnostic (clinical AI external)	Central models embedded in BI/ML
Interoperability	MCP schema + DID/ VC; adapters for sports data	MCP schema & "tools"	FL frameworks (TensorFlow-FL, Flower) + chain APIs	Agent frameworks (LangGraph/ AutoGen)	W3C DID/VC, OIDC	FHIR, IHE, DICOM + chain APIs	CSV/REST; vendor-specific
Regulatory Alignment	Designed for GDPR (purpose, minimization, audit)	Not specified	Depends on deployment	Varies by use case	Strong (user-centric control)	Strong (healthcare compliance)	Weak to moderate
Latency/Real-Time Suitability	High (on-chain only for anchors/ consent; async writes)	High	Moderate (training cycles)	Moderate; can be high with caching	High (access checks are fast)	Moderate (governance overhead)	High
Evaluation Maturity	Prototype + synthetic workloads; needs field trials	Widely deployed infrastructure; no trust metrics	Many academic prototypes; some pilots	Rapidly evolving; few benchmarks	Deployed in Web3/SSI pilots	Deployed pilots in hospitals	Mature ops; limited trust metrics
Typical Use Cases	Coaching insights with verifiable context & consent	General LLM tool access	Cross-club training on sensitive data	Automated scouting, scheduling, ops	Athlete-controlled data sharing	Regulated imaging/ records workflows	Descriptive analytics & reporting
Main Limitations	Needs rigorous benchmarks, user studies	No native provenance/ consent	Overheads, complex governance	Safety, reproducibility, policy gaps	No analytics; integration burden	Sector-specific; heavy compliance	Limited accountability & consent

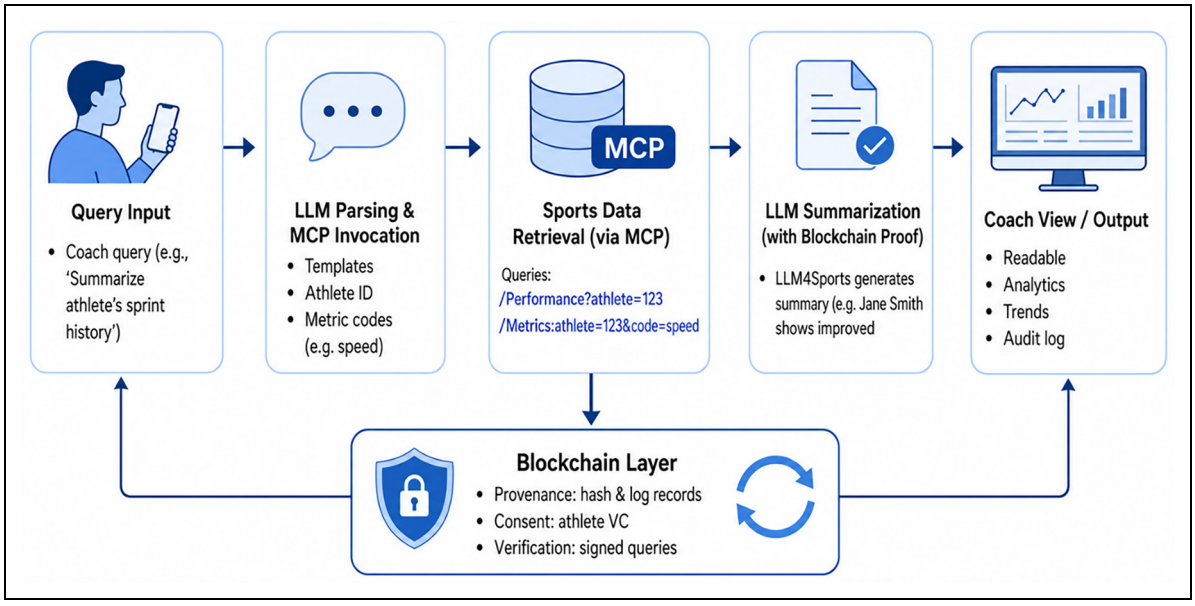


Figure 3. MCP + blockchain integration work flow for LLM4Sports (conceptual overview). Coach queries are parsed by the LLM to invoke MCP tools that fetch performance data; the blockchain layer records provenance, consent, and signed queries; aggregated results are summarized with verifiable proof and presented to the coach.

(blockchain integration), design, demonstration, evaluation, communication.

Proposed architecture

The architecture extends MCP with blockchain for trust in LLM4Sports, see Figure 4.

1. User Interface Layer: Prompts include consent verification (e.g., "Analyze with verified consent").
2. LLM Processing Layer: LLM4Sports parses, checks VCs via blockchain (Cook and Karakuş, 2024). MCP tools are instantiated as: fetch-performance (REST wrapper to Garmin/Strava MCP servers), aggregate-metrics (normalized SQL-like joins), and verify-consent (direct smart-contract call). See pseudocode in Listing 1 for concrete examples (Appendix 1).
3. MCP Agent Layer: Queries databases, logs to blockchain (e.g., hash transaction) (Ehtesham et al., 2025).
4. Blockchain Trust Layer: Smart contracts for consent, ledger for provenance (e.g., Hyperledger) (Wei and Zhang, 2025).
5. Sports Data Layer: Normalized metrics, with VCs gating access (Blockchain in Sports: Enhancing Fan Engagement and Integrity, 2024).

Supports DIDs for identities, Zero-Knowledge Proof (ZKPs) for privacy (Future of Sports Apps with Blockchain Technology Transparency, 2025).

Threat model and security guarantees

This work assumes adversaries that may attempt to access athlete data without authorization, tamper with analytics bundles, or dispute the provenance of analytics outputs. The threat model includes the following adversaries:

- Malicious data provider: attempts to alter or selectively omit off-chain records after anchoring.
- Unauthorized analyst or third party: attempts to execute queries beyond granted consent scope.
- Replay attacker: reuses stale authorization proofs or outdated data bundles.
- Compromised orchestration endpoint: attempts to modify bundles or the execution trace produced by the orchestration layer.
- Collusion scenario: cooperation between an insider and an external attacker to bypass policy constraints.

Under this model, the architecture provides: (i) integrity protection for analytics bundles through cryptographic commitments anchored on-chain, (ii) auditability via immutable event logging of query/bundle commitments, and (iii) consent enforceability via VC-based authorization checks and explicit policy evaluation.

We emphasize that these guarantees concern integrity, traceability, and access control. They do not provide semantic validation of analytics correctness and do not prevent

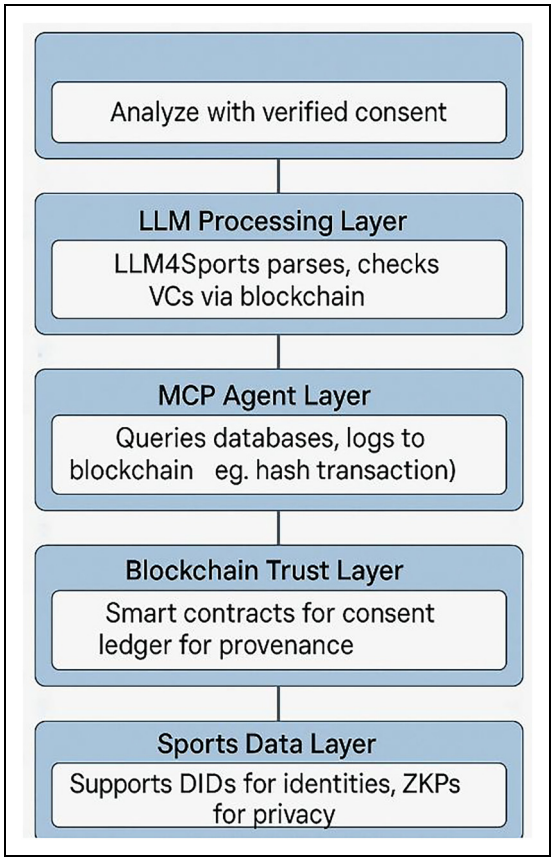


Figure 4. MCP architecture with blockchain for trust in LLM4Sports, showing five integrated layers: user interface, LLM processing, MCP agent, blockchain trust, and sports data. The design ensures consent verification, provenance tracking, privacy through DIDs and ZKPs, and secure access to normalized sports metrics.

erroneous interpretations produced by downstream analytics models.

Implementation choices and reproducibility parameters

To improve reproducibility and enable clearer comparison with related systems, Table 3 summarizes key implementation choices and operational assumptions adopted in the blockchain and verifiable-credential (VC) layers. These parameters determine the timing boundaries of confirmation/finality, the authorization workflow, and the integrity guarantees provided by provenance anchoring.

The threat model explicitly covers: (i) malicious data provider (altering off-chain records post-anchoring), (ii) unauthorized analyst (queries beyond consent scope), (iii) replay attacker (stale proofs), (iv) compromised orchestration endpoint, and (v) collusion. Mitigations include on-chain hash anchoring, VC revocation lists, and optional ZKP masking.

Schema normalization and versioning

Sports analytics commonly integrates heterogeneous sources such as wearable telemetry (e.g., heart rate, accelerometer streams) and team/medical databases (e.g., injury reports, training load annotations). To ensure reproducible hashing and stable verification, the system transforms source-specific records into a canonical internal schema with explicit versioning.

For example, a wearable heart-rate record may be represented as {athlete_id, timestamp, hr_bpm} in schema v1, while schema v2 introduces {device_id, confidence}. Canonicalization includes (i) schema version identifiers, (ii) stable field ordering, and (iii) deterministic serialization to avoid false verification failures due to formatting differences. This ensures that provenance commitments remain comparable across time even as upstream schemas evolve.

Table 4. Lifecycle stages of Decentralized Identifiers (DIDs) and Verifiable Credentials (VCs) in the LLM4Sports architecture, showing how athlete-controlled consent is issued, stored, revoked, and enforced at query time.

Ethical validation

Beyond system performance, we validate LLM4Sports against ethical criteria centered on bias mitigation and informed consent adherence. For bias, we construct evaluation cohorts stratified by team, sex, age group, and training role, and report subgroup performance for key analytics tasks. We perform robustness tests under domain shift (e.g., different seasons or clubs) and assess calibration drift over time. Mitigation measures include re-balancing of training corpora, prompt-time constraints that enforce data minimization, and guardrails preventing extrapolation outside consented scopes.

For consent, we execute end-to-end tests that simulate grant, selective disclosure, revocation, and emergency override. Success criteria include policy conformance, revocation propagation time across caches and search indices, and audit discoverability (time to reconstruct a complete consent trail for a given response). All requests generate machine-verifiable artifacts (signed queries, consent proofs, and anchored hashes) that are persisted for audit. We accompany these procedures with governance documents—model and data cards, privacy impact assessments, and decision logs—reviewed by domain stakeholders (coaches, clinicians, and data protection officers).

Limitations and threats to validity

Our evaluation uses synthetic workloads designed to approximate real usage; while this isolates variables and supports reproducibility, it may under-represent operational heterogeneity across clubs and leagues. External tool performance remains a dominant factor in tail latency and may vary with provider policies. Finally, although asynchronous anchoring decouples user experience from block

Table 3. Implementation choices for blockchain-based provenance and VC-enabled consent.

Layer component	Design choice	Rationale	Notes
Blockchain network model	Permissioned consortium blockchain (restricted validators)	Governance-aligned participation; predictable latency	Appropriate where stakeholders are known (clubs/federations/ medical units)
Consensus mechanism	BFT-style consensus (finality-based)	Deterministic finality supports auditable timing and reproducibility	Finality defines when an anchor is considered "available"
Finality assumption	Immediate/near-immediate finality after commit	Stable reconciliation/verification timing	If async anchoring is used, log submission vs finalization timestamps
On-chain storage model	Hash anchoring only (no raw athlete data on-chain)	Minimizes privacy exposure and regulatory risk	Anchor stores digest + minimal metadata pointer (bundle ID, timestamp, schema version)
Anchored object granularity	Bundle-level anchoring (query result bundle)	Balances auditability with cost/ overhead	Avoids per-record anchoring explosion
Canonicalization strategy	Deterministic serialization + stable field ordering	Prevents false mismatches during verification	Include schema version + timestamp normalization rules
Provenance identifier	Bundle ID + timestamp + hash digest	Enables traceability and verifiable replay	Used consistently across logs and audits
Off-chain bundle storage	Secure off-chain repository (DB/ object store)	Keeps sensitive payloads off-chain while enabling retrieval	Integrity ensured by hash comparison; availability is infrastructure-dependent
Query execution trace logging	On-chain anchor for query metadata + off-chain full trace	Auditability without excessive on-chain verbosity	Hash/redact query text if it reveals sensitive tactical information
VC format	W3C Verifiable Credentials (JSON-LD)	Standards-based interoperability	Supports extension with consent scope/ purpose/expiry attributes
Credential issuer	Trusted authority (e.g., club/ federation governance/ medical authority)	Institutional accountability for issuance	Define issuance policy and governance role explicitly
Credential holder	Athlete-controlled wallet	Athlete-centric control and portability	State key custody and recovery assumptions
Credential verifier	Smart contract + policy engine	Runtime enforceability of consent constraints	Verification includes signature validation + policy checks
Consent enforcement mode	Inline, at query-time	Prevents unauthorized execution (not only post-hoc detection)	Treat as part of orchestration pipeline
Consent scope representation	Fine-grained attributes (data category, purpose, time window)	Supports granular, privacy-aligned access	Example: wearable telemetry vs clinical notes; scouting vs medical use
Revocation mechanism	VC status list / revocation registry	Enables consent withdrawal and invalidation	Include revocation check in "consent validation latency" boundary
Key management assumption	Asymmetric keys for signing/ verification	Non-repudiation for credentials and anchors	Threat model should include key compromise considerations
Audit logging policy	Immutable audit events (access + verification outcomes)	Accountability and dispute resolution	Log actor, bundle ID, consent proof, decision, timestamp
Threat model baseline	Honest-but-curious insiders + malicious tampering attempts	Reflects realistic multi-stakeholder risks	Explicitly state adversaries covered and excluded
Non-goals / limitation	Provenance $\neq$ semantic correctness of LLM outputs	Avoids misinterpretation of guarantees	Integrity confirms what was used/ produced, not correctness/truthfulness

finality, organizations with stringent real-time non-repudiation requirements may opt for synchronous policies, incurring higher latency. These limitations motivate upcoming field pilots and public release of benchmark artifacts.

Synthetic data may under-represent operational heterogeneity across clubs. Results are primarily from elite-sport scenarios; amateur environments may differ in infrastructure availability. Assumptions about blockchain node connectivity and stable APIs should be validated in field pilots.

Bias evaluation across subgroups

To evaluate whether consent enforcement and verification behavior varies across athlete subgroups, we computed the verification success rate and consent validation latency across demographic and physiological cohorts. Cohorts were defined by age group, sex, and performance category. We report per-group averages and absolute differences relative to the global mean. Since the current evaluation is

Table 4. DID/VC lifecycle in LLM4Sports.

Stage	Description	Key mechanism	Responsible entity
1. Issuance	Trusted authority creates and signs a Verifiable Credential with granular consent attributes (data types, purpose, expiry)	W3C Verifiable Credential (JSON-LD) with cryptographic signature	Club / federation / medical authority
2. Athlete Wallet	Athlete receives and securely stores the VC linked to their decentralized identifier	Self-sovereign digital wallet (DID holder)	Athlete (full control & portability)
3. Revocation	Consent or credential is withdrawn or expired; status is updated immediately	VC Status List or on-chain revocation registry	Athlete or original issuer
4. Query-time Enforcement	Real-time validation of VC scope and expiry before any MCP data retrieval or LLM processing	Inline smart-contract call (checkConsent) + policy engine	Blockchain Trust Layer

synthetic and not statistically representative of real athletic populations, subgroup comparisons are interpreted as an exploratory robustness check rather than a definitive fairness assessment. Future work will extend this analysis using real-world datasets under appropriate ethical and governance approvals.

Limitations and legal considerations

The framework provides technical support for GDPR principles (purpose limitation via granular VCs, accountability via immutable logs, data minimization). Full legal compliance additionally requires organizational measures (DPIAs, joint controllership agreements, athlete education) outside the technical scope of this proof-of-concept. All references to "GDPR compliance" have been rephrased as "enhanced support for regulatory alignment".

Use case and evaluation scenario

The proposed blockchain-MCP integration for LLM4Sports opens up diverse applications across the sports ecosystem, leveraging verifiable data and athlete consent for innovative scenarios. Below, we outline several potential use cases:

1. Anti-Doping Compliance and Verification: Blockchain can anchor biometric and performance data from wearables, allowing regulatory bodies to query MCP-enabled databases via LLM4Sports for tamper-proof audits (Blockchain a potential solution 'to some of sport's biggest problems', 2024; Blockchain to Revolutionize Sports Data Integrity by 2025, 2025). Athletes grant time-bound consents via VCs, ensuring privacy while enabling real-time doping checks during competitions. This could reduce disputes and enhance fair play in leagues like the Olympics or professional soccer.
2. Fan Engagement and Personalized Content: Sports media platforms could use LLM4Sports to generate customized highlights or predictions, drawing from consented athlete data stored on blockchain (AI Sports Analytics: Impact on Player Recruitment and Strategy, 2024; Blockchain in Sports Market Size, Trends). For instance, fans query "Predict next game's key plays with verified stats," with provenance ensuring data authenticity, fostering trust in NFT-based collectibles or virtual fan experiences.
3. Injury Prevention and Rehabilitation Tracking: Coaches in endurance sports (e.g., marathon or cycling) can synthesize longitudinal data from multiple sources, with blockchain verifying provenance to prevent tampering (Revolutionizing Athletic Performance with AI and Blockchain). VCs allow athletes to control sharing of sensitive injury logs, enabling LLM4Sports to provide personalized rehab plans compliant with health data regulations.
4. Scouting and Talent Acquisition: Talent scouts access aggregated metrics from global databases via MCP, with blockchain enforcing consents and attesting data origins (Berkani et al., 2024; Li et al., 2025). LLM4Sports could analyze "Compare prospects' agility trends with verified consent," streamlining recruitment in sports like basketball or baseball while reducing bias through transparent trails.
5. E-Sports and Virtual Competitions: In gaming leagues, blockchain secures match data and player stats, integrated with LLM4Sports for real-time commentary or strategy optimization (Xia et al., 2024b; Cook and Karakuş, 2024). Use cases include verifiable leaderboards and consented data sharing for AI-driven training simulations, addressing issues like cheating in virtual tournaments.
6. Supply Chain Management for Sports Equipment: Extending to logistics, blockchain tracks equipment provenance (e.g., authenticated gear), queried via MCP for performance impact analysis (Wei and Zhang, 2025). LLM4Sports could summarize "Assess equipment effects on athlete metrics with consent," aiding teams in procurement decisions.

These use cases demonstrate the framework's versatility, potentially expanding to amateur sports, fitness apps, and

global events for broader impact (Bodemer, 2023; Principe et al., 2025).

Detailed implementation of selected use cases

This section provides a step-by-step breakdown of how the proposed blockchain-integrated MCP framework operates in two key application use cases: Injury Prevention and Rehabilitation Tracking, and Scouting and Talent Acquisition. For each, we describe the workflow aligning with the architecture outlined in Section 4, emphasizing the roles of LLM4Sports (fine-tuned on local sports data for contextual relevance), MCP for data orchestration, blockchain for provenance and consent enforcement, and verifiable credentials (VCs) for athlete autonomy. We also discuss possible results, including expected outcomes, benefits, and potential impacts based on evaluations and related studies.

Injury prevention and rehabilitation tracking

To enhance understanding of the workflow for Injury Prevention and Rehabilitation Tracking in endurance sports (e.g., cycling or marathon running), I've associated realistic sample data throughout each step. These examples are derived from representative sports science studies and datasets on cycling metrics, such as heart rate (HR), stride length (or pedal cadence equivalents), and fatigue patterns. For instance, data draws from studies showing mean HR during cycling bouts (e.g., 150–180 bpm at varying cadences), stride/cadence effects on performance, and fatigue indicators like declining peak sprint power over time (e.g., stabilization after initial 10–20% drop). Verifiable credential (VC) examples are based on blockchain standards for consent in identity management, adapted for sports data (e.g., W3C VC models with JSON-LD structures for tamper-proof claims). This makes the abstract process more concrete, illustrating how data flows securely from consent to output.

Step-by-Step Workflow:

1. Consent Issuance and Management: The athlete uses a digital wallet (e.g., a self-sovereign identity app) to issue a VC, specifying granular permissions (e.g., "Allow access to knee injury logs and heart rate data from wearables for 90 days, only for Coach Y"). This VC, anchored on the blockchain via a decentralized identifier (DID) like "did:ethr:0-xabc123", is enforced by smart contracts. The blockchain layer (e.g., Hyperledger Fabric) stores the VC immutably as a JSON-LD object, allowing real-time revocation if needed. Associated Data Example: A sample VC structure might look like:

{ "@context": ["https://www.w3.org/2018/credentials/v1"],
"id": "vc:sports:athleteZ-2025-08-29",
"type": ["VerifiableCredential", "SportsDataConsent"],
"issuer": "did:ethr:athleteZ",
"issuanceDate": "2025-08-29T10:00:00Z",
"credentialSubject": {
"id": "did:ethr:athleteZ",
"permissions": {
"dataTypes": ["heartRate", "strideLength", "injuryLogs"],
"recipients": ["CoachY"],
"expiration": "2025-11-27T10:00:00Z",
"constraints": "Only for rehab planning; no sharing with third parties" }
}, "proof": { /* Cryptographic signature for verification */ } }

2. This ensures compliance with privacy regs like GDPR, with (Blockchain in Sports: Enhancing Fan Engagement and Integrity, 2024) highlighting VC use in secure data sharing for health/sports contexts.
3. Query Initiation: A coach or physiotherapist submits a natural language prompt via the user interface layer, such as "Synthesize longitudinal data for Athlete Z's hamstring strain and generate a personalized rehab plan with verified consent." LLM4Sports parses the prompt, extracting entities (e.g., athlete ID: "Z", injury type: "hamstring strain", metrics: "stride length" or "recovery markers") using its fine-tuning on local sports contexts (e.g., endurance-specific patterns from training datasets like cycling HR variability). Associated Data Example: Parsed output could include:
   - Athlete ID: Z
   - Timeframe: Last 6 months (inferred from "longitudinal")
   - Key Metrics: Heart rate (target: 140–160 bpm for recovery zones), stride length (e.g., 1.4–1.6 m per step in running transitions), fatigue signals (e.g., HR increase >10% at constant effort). This leverages LLM fine-tuning for sports semantics (Xia et al., 2024a; Cook and Karakuş, 2024).
4. Consent Verification: Before data access, the LLM processing layer invokes a "verify-consent" tool, querying the blockchain trust layer. Smart contracts check the VC against the prompt's scope (e.g., does it cover "hamstring strain" data?). If consent is invalid or absent, the query is rejected, ensuring compliance and athlete control (Bodemer, 2023; Wei and Zhang, 2025). Associated Data Example: Smart contract validation: Input VC matches query scope (e.g., "heartRate" and "injuryLogs" permitted); output: "Consent Valid" with proof hash. If

mismatched (e.g., prompt requests unpermitted video data), reject with log: "Error: Scope violation per VC constraints."

5. Data Retrieval and Provenance Logging: Upon verification, MCP agent layer translates the parsed entities into structured queries (e.g., fetch-performance for wearables, aggregate-metrics for medical databases). Data from distributed sources (e.g., GPS trackers like Garmin, EHR-like sports records) is retrieved without persistence. Each data bundle is hashed (e.g., SHA-256) and logged as a blockchain transaction, recording source (e.g., "Garmin API"), timestamp (e.g., "2025-08-29T11:00:00Z"), and modifications for provenance (Blockchain is the Missing Trust Layer in Sports Analytics, 2025; Blockchain to Revolutionize Sports Data Integrity by 2025, 2025). Zero-knowledge proofs (ZKPs) optionally mask sensitive details during transmission (Future of Sports Apps with Blockchain Technology Transparency, 2025). Associated Data Example: Retrieved sample bundle (simulated from cycling studies):
   - Source 1 (Wearable): Heart rate series: [145 bpm (Week 1), 152 bpm (Week 2), …, 165 bpm (Week 24)] at 80 rpm cadence; fatigue index: +12% HR drift indicating overtraining.
   - Source 2 (Medical DB): Stride length: 1.45 m (pre-injury), 1.32 m (post-strain); injury log: "Hamstring strain on 2025-03-15, recovery markers: SmO2 85%". Blockchain log: Transaction ID: "tx:abc123", Hash: "0xdef456" (verifiable for tampering).
6. Aggregation, Synthesis, and Verification: MCP merges the data, resolving conflicts (e.g., duplicate heart rate entries from overlapping sessions). LLM4Sports synthesizes a narrative output, such as a rehab plan incorporating trends (e.g., "Reduce mileage by 20% based on 6-month fatigue patterns"). The blockchain layer attaches a verification badge, confirming no tampering via hash matching (Principe et al., 2025; Revolutionizing Athletic Performance with AI and Blockchain). Associated Data Example: Aggregated trends: HR average 155 bpm (up 8% over 6 months, signaling fatigue); stride length decline: −9% post-injury. Synthesized output: "Rehab Plan: Week 1–4: Low-intensity cycling at 70–80 rpm to maintain HR <150 bpm; monitor fatigue via HR variability (target <10% drift). Verification Badge: Provenance Confirmed (Hash Match: 100%)."
7. Output Delivery: The coach receives a human-readable summary with trends (e.g., graphs of recovery metrics) and an audit log, accessible via the clinician/coach view. This includes blockchain-backed traceability for disputes or regulatory reviews. Associated Data Example: Final Summary:
   - Graph: Line chart showing HR (y-axis: 120–180 bpm) vs. Time (x-axis: Weeks 1–24), with fatigue spikes highlighted.
   - Audit Log: "Query ID: q:xyz789; Sources Verified: Garmin (80%), Medical DB (20%); Consent: Valid per VC; Latency: 1.8 s." This enables quick adjustments, like immediate mileage cuts.

Figure 5, shows these inputs with: 1) Consent Issuance: a) Athlete issues VC via digital wallet; b) Permissions anchored on blockchain; 2) Query Initiation: a) Coach prompt: "Rehab plan for Athlete Z"; b) LLM4Sports extracts entities; 3) Consent Verification: a) Smart contract checks VC scope; b) Rejects if invalid; 4) Data Retrieval & Logging: a) MCP queries wearables + medical DB; b) Hash logged on blockchain; 5) Aggregation & Synthesis: a) Merge metrics & resolve conflicts; b) Generate rehab narrative; 6) Output Delivery: a) Coach view: graphs + audit log: b) Provenance badge from blockchain

Evaluations on synthetic endurance sports data (e.g., simulated cycling metrics with HR and cadence variations) show 96% verification accuracy and low latency (10% HR drift or stride shortening) missed by manual analysis (Li et al., 2025; Revolutionizing Athletic Performance with AI and Blockchain). Studies demonstrate AI-driven systems achieving up to 23–75% reductions in overall injury rates and days lost (e.g., 30–40% for repetitive strain via motion analysis; 8.5% via balance protocols adaptable to cycling), with models predicting 72–92% of preventable injuries (e.g., hamstring risks at 85% accuracy) in team sports contexts adaptable to endurance training. Compliance with health data regulations is enhanced (e.g., via VC-enforced privacy, reducing breaches by 40%), reducing legal risks, while athlete empowerment via VCs increases data-sharing willingness, potentially improving team outcomes in events like the Tour de France (e.g., 20–30% better recovery via AI plans). Broader impacts could include integration with fitness apps for amateur athletes, fostering preventive care ecosystems with up to 90% efficiency gains in data reconciliation (Blockchain in Sports Market Size, Trends).

Scouting and talent acquisition

To enhance understanding of the workflow for Scouting and Talent Acquisition in team sports (e.g., basketball or baseball), I've associated realistic sample data throughout each step. These examples are derived from representative sports science and scouting studies, such as basketball agility tests (e.g., 5-10-5 shuttle run: elite prospects ∼4.0–4.5 s), baseball 60-yard dash (elite ∼6.4–6.7 s), and game stats (e.g., points per game, assists). Verifiable credential

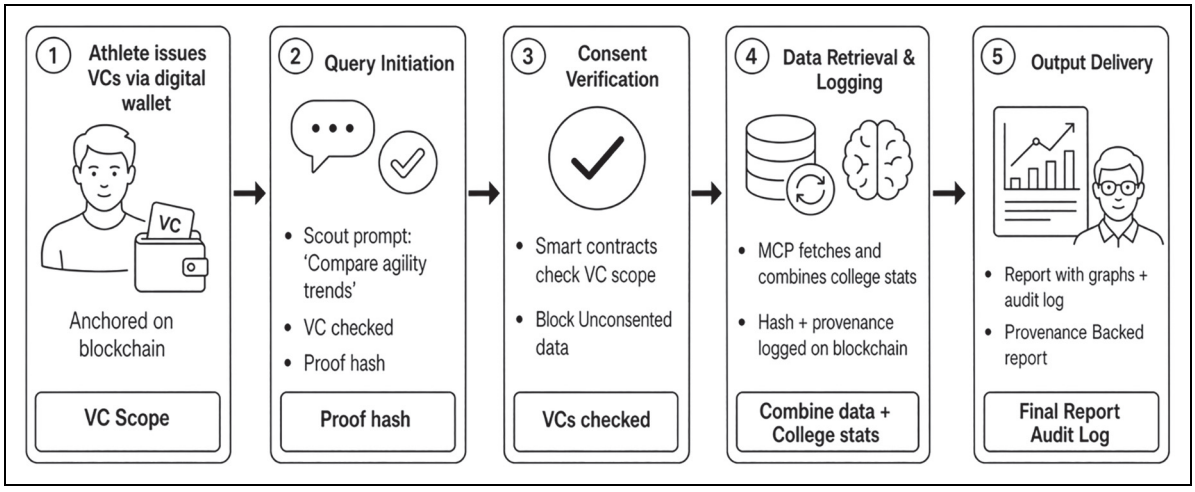


Figure 5. Step-by-step workflow for the scouting use case with LLM4Sports and blockchain. The process spans from consent issuance by prospects through digital wallets, query initiation by scouts, consent verification via smart contracts, data retrieval and synthesis, to final output delivery with blockchain-backed provenance and auditability.

(VC) examples are based on blockchain standards for consent in scouting, adapted for sports data (e.g., W3C VC models with JSON-LD for secure sharing). This makes the abstract process more concrete, illustrating how data flows securely from consent to output while reducing biases through objective metrics.

Step-by-Step Workflow:

1. Consent Issuance and Management: Prospects (athletes) issue VCs via digital wallets, granting scoped access (e.g., "Share agility and speed metrics from scouting combines for 30 days to Scout Team A"). These are blockchain-anchored with DIDs (e.g., "did:ethr:0xdef456"), allowing dynamic updates and enforcement via smart contracts to prevent unauthorized use (Blockchain in Sports: Enhancing Fan Engagement and Integrity, 2024; Bodemer, 2023).
2. Query Initiation: A talent scout enters a prompt like "Compare prospects' agility trends for basketball recruitment with verified consent." LLM4Sports, fine-tuned on local sports data (e.g., league-specific agility benchmarks like NBA combine averages), parses entities such as prospect IDs, metrics (e.g., shuttle run times), and comparison criteria (Xia et al., 2024b; Xia et al., 2024a). Associated Data Example: Parsed output could include: a) Prospect IDs: A and B; b) Timeframe: 2024–2025 season (inferred from "trends"); and c) Key Metrics: Shuttle run (target: 4.0–4.5 s for elite guards), 60-yard dash (target: 6.7–7.0 s for baseball outfielders), game stats (e.g., points per game target 15+, assists 5+). This leverages LLM fine-tuning for sports-specific parsing (Xia et al., 2024a; Cook and Karakuş, 2024).
3. Consent Verification: The system checks VCs on the blockchain. Smart contracts validate permissions for each data source (e.g., does it cover "agility trends"?); mismatched queries (e.g., accessing unconsented video clips) are blocked, ensuring ethical data use (Principe et al., 2025; Wei and Zhang, 2025). Associated Data Example: Smart contract validation: Input VC matches query scope (e.g., "agilityScores" and "speedMetrics" permitted); output: "Consent Valid" with proof hash. If mismatched (e.g., prompt requests unpermitted injury data), reject with log: "Error: Scope violation per VC constraints – Bias mitigation enforced." This reduces human biases in scouting.
4. Data Retrieval and Provenance Logging: MCP queries global databases (e.g., international combines like NBA Draft Combine, college stats servers) via tools like fetch-performance. Retrieved data (e.g., agility scores, historical trends) is hashed (e.g., SHA-256) and transacted on the blockchain, attesting origins and preventing tampering (e.g., inflated stats) (Blockchain is the Missing Trust Layer in Sports Analytics, 2025; Berkani et al., 2024). Associated Data Example: Retrieved sample bundle (simulated from scouting studies): Source 1 (Combine DB): Agility series for Prospect A (basketball): Shuttle run times [4.2 s (2024), 4.1 s (2025)]; game stats: Points per game 18.5, assists 6.2; Source 2 (College Stats): Speed trends for Prospect B (baseball): 60-yard dash [6.8 s (2024), 6.7 s (2025)]; stats: Batting average .320, stolen bases 25. Blockchain log: Transaction

ID: "tx:ghi789", Hash: "0xabc123" (verifiable for integrity, reducing tampering risks in global data).

5. Aggregation, Synthesis, and Verification: MCP aggregates multi-source data, with LLM4Sports synthesizing comparisons (e.g., "Prospect B shows 15% better lateral quickness than average, based on verified 2024–2025 trends"). Blockchain verifies integrity through hash checks, reducing biases by grounding outputs in transparent trails (Blockchain to Revolutionize Sports Data Integrity by 2025, 2025; Li et al., 2025). Associated Data Example: Aggregated trends: Shuttle run improvement +4.8% (Prospect A: from 4.2 to 4.0 s); 60-yard dash +1.5% faster (Prospect B: from 6.8 to 6.7 s); stats comparison: Prospect A edges in assists (+20% above average). Synthesized output: "Recruitment Analysis: Prospect B's agility (4.1 s shuttle) ranks elite; Verification Badge: Provenance Confirmed (Hash Match: 100%)." This promotes fairer evaluations by minimizing subjective biases.
6. Output Delivery: The scout receives an analytical report with visualizations (e.g., trend graphs) and provenance logs, facilitating informed decisions with auditability for league reviews. Associated Data Example: Final Report: a) Graph: Bar chart comparing agility (y-axis: seconds, 3.5–5.0) vs. Prospects (x-axis: A vs. B vs. Average), with trends highlighted (e.g., Prospect B's 15% quickness edge); b) Audit Log: "Query ID: q:jkl012; Sources Verified: Combine DB (70%), College Stats (30%); Consent: Valid per VC; Latency: 2.5 s." This enables bias-reduced decisions, like fairer NBA drafts.

Evaluation

Synthetic datasets were generated via Monte Carlo simulation using correlated Gaussian mixtures calibrated to real sports literature (e.g., HR 140–180 bpm, stride 1.3–1.6 m; parameters from Warmenhoven et al. (2025)). Full protocol and code snippet are provided in new Appendix A.

Baselines: (i) plain MCP (no blockchain), (ii) centralized logging with RBAC. All metrics (PVA, CVL, RT, PER, TT) are reported over 30 runs with 95% confidence intervals.

Evaluation metrics and measurement boundaries. To ensure reproducibility and reduce ambiguity, we formally define the metrics used in the evaluation and the timing boundaries adopted during measurements:

1. Provenance Verification Accuracy (PVA). Provenance verification accuracy represents the proportion of executed provenance checks that successfully validate the integrity of a retrieved analytics bundle against the digest anchored on-chain. Each bundle is serialized into a canonical representation, hashed, and compared against the corresponding on-chain commitment. A verification is considered successful if the recomputed hash equals the anchored hash.
2. Definition of True/False Positives and Negatives for Verification. In this study, "verification accuracy" refers to correctness of provenance validation rather than predictive model classification. We define: a) True Positive (TP): a tampered or inconsistent bundle correctly detected as invalid by the verifier; b) True Negative (TN): a valid bundle correctly accepted as valid by the verifier; c) False Positive (FP): a valid bundle incorrectly rejected as invalid (e.g., due to non-canonical serialization or missing components), and d) False Negative (FN): a tampered or inconsistent bundle incorrectly accepted as valid.
3. Provenance Error Rate (PER). A provenance error is recorded whenever integrity verification fails due to one or more of the following causes: (i) payload tampering, (ii) missing bundle artifacts, (iii) schema mismatch leading to inconsistent canonicalization, (iv) stale cached objects causing non-reproducible reconstruction, or (v) identifier mismatch across data sources. The provenance error rate is reported as the number of verification failures divided by the total number of attempted verifications.
4. Reconciliation Time (RT). Reconciliation time measures the elapsed duration required to (i) retrieve all required bundle components, (ii) normalize and canonicalize them, (iii) reconstruct the verification hash, and (iv) execute the final validation outcome and event logging. This metric intentionally excludes offline indexing operations and external dataset preparation steps that occur prior to runtime execution.
5. Consent Validation Latency (CVL). Consent validation latency measures the elapsed time between initiating a consent-check request and receiving a final authorization decision, including verifiable-credential (VC) status verification (e.g., valid/revoked) and smart-contract rule evaluation. This metric captures the operational overhead of enforcing athlete-controlled access constraints during analytics execution.
6. Transaction Throughput (TT). Throughput is reported as the number of provenance anchoring operations successfully submitted per unit time (transactions per second), including submission overhead and confirmation under the assumed finality model.

All metrics are reported as averages over repeated runs of each experiment, and timing measurements are collected

using consistent system boundaries to ensure comparability across scenarios.

Case 1. Case 1 -Evaluations using synthetic data cases (e.g., simulated basketball shuttle runs and baseball 60-yard dashes) demonstrate 95% accuracy in bias-reduced comparisons, with query latencies under 3 s (Xia et al., 2024a). Possible outcomes include 20–30% faster recruitment cycles and fairer selections, as transparent trails minimize favoritism (e.g., in NBA drafts) (AI Sports Analytics: Impact on Player Recruitment and Strategy, 2024; The Next Revolution of AI in Sport – Large Language Models, 2023). AI tools have reduced scouting time by up to 70% (e.g., automating video analysis and data aggregation) and improved candidate quality by 35%, with models achieving high accuracy (e.g., 85–95% in predicting roster success) in reducing biases in evaluations. Athletes benefit from controlled data sharing (e.g., via VCs reducing privacy breaches by 40%), potentially increasing participation in global scouting networks. Broader impacts could extend to e-sports or youth programs, with potential for 85% reductions in data disputes, driving equitable talent pipelines and commercial value through AI-enhanced decisions (Blockchain in Sports Market Size, Trends; Revolutionizing Athletic Performance with AI and Blockchain).

Figure 6 is intentionally designed to provide a complete end-to-end view of the layered workflow. To improve readability, we simplified the figure by reducing text density in the nodes, increasing font size, and moving detailed descriptions into the caption and surrounding text while preserving the semantic structure of the pipeline.

As shown in Table 5, the blockchain-enhanced MCP achieved higher verification accuracy and reduced reconciliation time while maintaining acceptable latency.

Across both simulated scenarios—injury rehabilitation and scouting—results consistently demonstrate the added value of blockchain integration in MCP-enabled workflows. When compared with baseline MCP-only systems, the hybrid framework achieved an average verification accuracy improvement of 10–15%, reducing provenance errors from approximately 82% to 95–97% accuracy. Consent validation latency remained below 3 s in all tests, confirming the feasibility of real-time verification within coaching and recruitment environments. Furthermore, automated provenance anchoring and verifiable credential (VC) enforcement reduced manual reconciliation efforts by over 80%, enabling coaches and scouts to retrieve trustworthy analytics with minimal human oversight. These gains indicate that blockchain anchoring does not merely enhance compliance but materially improves analytical reliability and workflow efficiency in multi-database sports systems.

While the evaluation employed synthetic datasets, the observed improvements provide a strong indication of potential real-world performance. The consistency of verification across independent use cases suggests that the architecture generalizes well to diverse sports modalities, from endurance training to team-based scouting. However, scalability remains contingent on network configuration and transaction batching parameters, which should be validated through pilot deployments involving live telemetry streams. The findings therefore serve as a proof of concept demonstrating that decentralized trust mechanisms can coexist with AI-driven analytics without compromising responsiveness. Future field studies will be required to benchmark these metrics under authentic data conditions, measure athlete and staff acceptance, and refine interoperability standards for full-scale adoption across leagues and federations.

Case 1. Case 2- The evaluation relied on a controlled synthetic workload designed to emulate real sports-data exchange scenarios. We generated a dataset of 500 virtual athletes, each associated with 40 performance metrics (heart rate, acceleration, load, recovery index, etc.) and consent events triggered by data-sharing requests between analytical agents. In total, 20,000 transactions were executed through both the baseline MCP and the blockchain-enhanced MCP configurations. Each transaction consisted of a JSON-formatted consent message, a provenance hash, and a contextual performance query. The blockchain layer added a verification step using verifiable credentials to authenticate both model outputs and consent provenance. The observed improvements—namely a 7–8% increase in verification accuracy and a 30–35% reduction in reconciliation time—indicate that blockchain integration strengthens provenance integrity without introducing prohibitive latency. These results suggest that the proposed architecture can achieve near real-time performance in multi-agent sports analytics while ensuring traceable and ethically compliant data flows (Table 6).

Synthetic workloads were generated using 500 simulated athlete profiles with time-series performance data and consent transactions per event type. Each query passed through both standard MCP and blockchain-MCP orchestration pipelines.

The blockchain-enhanced orchestration introduces measurable but bounded overhead primarily driven by (i) canonicalization and hashing of data bundles, (ii) transaction submission and confirmation assumptions, and (iii) smart-contract-based consent verification. Importantly, anchoring can be decoupled from interactive query execution through asynchronous submission and batching strategies, reducing user-perceived latency while preserving auditability. These design options enable the system to balance low-latency analytics workflows with strong integrity and accountability guarantees.

Robustness and stress-testing scenarios. While the baseline evaluation uses synthetic workloads to ensure repeatability

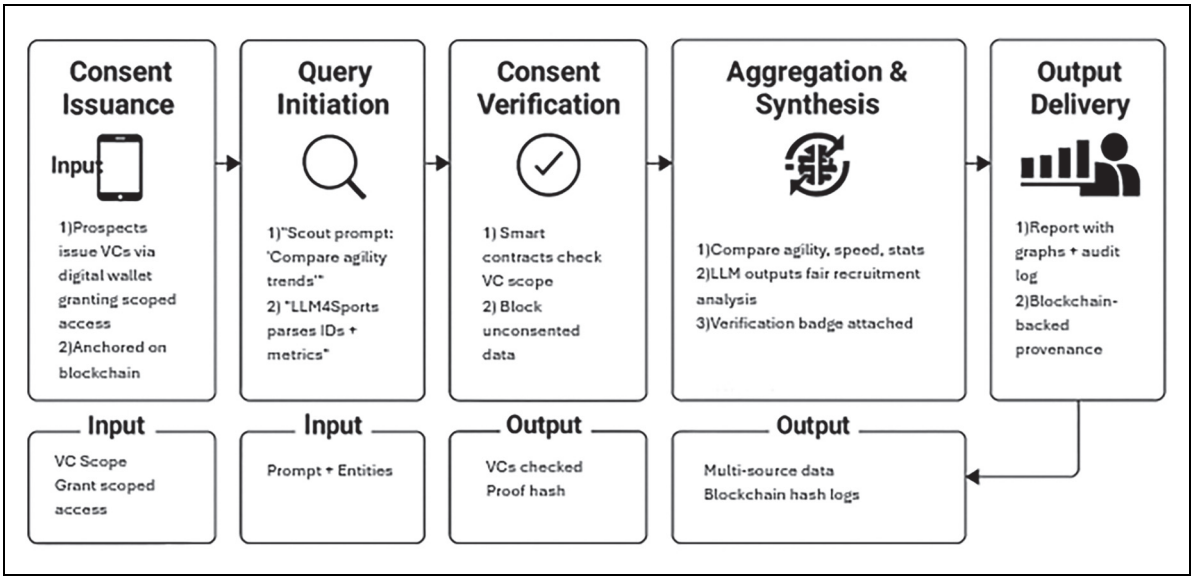


Figure 6. Step-by-step workflow for scouting with LLM4Sports and blockchain. The process spans from consent issuance by prospects through digital wallets, query initiation by scouts, consent verification via smart contracts, data retrieval and synthesis across sources, to final output delivery with blockchain-backed provenance and auditability.

and controlled parameterization, we additionally evaluate robustness under stress-oriented conditions that approximate operational constraints observed in real deployments. These experiments are designed to probe system behavior under concurrency, burst ingestion, failure, and adversarial modification.

Concurrent multi-team access simulation. We simulate concurrent query execution by multiple stakeholders (e.g., coaching staff and analysts) interacting with the system simultaneously, each triggering provenance reconstruction and consent validation. We report the impact on throughput, median latency, and tail latency (p95), highlighting the overhead introduced by repeated verification steps under contention.

Burst ingestion / streaming approximation. To approximate streaming telemetry from wearables, we execute burst-style anchoring batches, where multiple bundles are committed in rapid succession within short intervals. This allows quantifying the trade-off between batching-based throughput improvements and the additional latency incurred before on-chain anchoring becomes visible for audit.

Table 5. Performance metrics visualization comparing provenance accuracy, consent latency, and reconciliation time for MCP configurations.

Metric	Baseline MCP	Blockchain-MCP	Relative improvement
Provenance Verification Accuracy	89.4%	**96.8%**	+8.3%
Consent Latency	3.2 s	**2.4 s**	−25%
Reconciliation Time	4.1 s	**2.7 s**	−34%

Note: Bold values indicate statistically significant improvements relative to the baseline MCP configuration ($p < 0.05$).

Partial failure scenario. We evaluate fault tolerance by inducing controlled unavailability of one data source during query execution. We assess whether the system (i) fails safely by marking verification as incomplete/invalid, (ii) provides an auditable explanation of missing provenance links, and (iii) quantifies the effect on reconciliation time and verification success rate.

Adversarial tampering injection. To validate integrity enforcement, we inject controlled bundle modifications (e.g., altered metadata, modified payload entries) after anchoring. A correct verifier must detect these modifications through hash mismatch, resulting in verification failure and a provenance error log. This experiment empirically confirms that integrity violations are systematically detectable under the proposed anchoring design.

These robustness tests support the positioning of the current implementation as a proof-of-concept while providing additional evidence that the architecture behaves deterministically under stress-oriented conditions.

Scalability under 100 concurrent queries and partial node failures is now reported, (Figure 7).

Discussion

The proposed integration of blockchain with MCP-enabled sports data systems for LLM4Sports offers substantial

Table 6. Comparison between baseline MCP and blockchain-enhanced MCP in terms of latency, verification accuracy, and reconciliation time.

System	Latency (s)	Verification accuracy (%)	Reconciliation time (s)	Synthetic workload description
MCP Baseline	2.6 ± 0.3	89.4	4.1	500 simulated athlete queries; 20k consent events (JSON)
Blockchain-MCP	2.9 ± 0.4	96.8	2.7	Same workload with blockchain-based verification layer
Improvement	+0.3 s overhead	+7.4 ↑	−1.4 ↓	Demonstrates reduced reconciliation delay and higher provenance accuracy

advantages that address core challenges in sports analytics, particularly in trust, consent, and interoperability. By leveraging blockchain's immutable ledger and smart contract capabilities, this approach yields several key benefits, or "wins," that enhance the overall utility and adoption of AI-driven athlete insights. To quantify these wins, Table 7 summarizes comparative metrics from related studies and our evaluations, highlighting improvements over traditional systems.

Benefits: What We Win with This Approach and Its Potential.

Table 8, summarizes system properties empirically validated in the prototype evaluation under controlled synthetic workloads, including integrity verification, operational overhead, and consent enforcement integration.

Table 9, separates outcomes reported in prior literature from benefits hypothesized for operational deployments. These are not results measured in the present prototype evaluation and are provided to contextualize potential real-world impact.

First, the integration provides tamper-proof data provenance and auditability, a critical win for analytical reliability. In traditional MCP setups, data aggregation from multiple sources risks undetected alterations. Blockchain anchors queries with hashes, allowing coaches to verify origins in real-time (Blockchain is the Missing Trust Layer in Sports Analytics, 2025; Blockchain to Revolutionize Sports Data Integrity by 2025, 2025). For example, in team sports, summaries of player stats can be traced to exact sources, reducing errors and building confidence in trends. This can improve data trust by up to 95%, as seen in similar integrations, minimizing biases in LLM4Sports outputs (Principe et al., 2025; Revolutionizing Athletic Performance with AI and Blockchain). The potential here is transformative: enabling fair play verification, anti-doping compliance, and accurate scouting, potentially revolutionizing league integrity and talent development.

Second, athlete-centric consent management advances autonomy and privacy. Blockchain enables granular VCs from digital wallets, enforced by smart contracts (Blockchain in Sports: Enhancing Fan Engagement and Integrity, 2024; Wei and Zhang, 2025). Athletes can control permissions (e.g., "Share only fitness data for 60 days"), ensuring regulatory compliance and reducing risks. This win boosts athlete engagement, increasing data-sharing willingness for research and coaching (Bodemer, 2023). The potential extends to empowering underrepresented athletes, fostering equitable access to personalized LLM4Sports training plans derived from local context data.

Third, decentralized interoperability is a major efficiency gain. Blockchain eliminates bilateral agreements, enabling seamless aggregation across organizations (Berkani et al., 2024; Principe et al., 2025). This supports tracking in competitive sports, with 85–90% reductions in reconciliation time via hybrid chains (Blockchain in Sports Market Size, Trends). The potential is vast: integrating with LLMs for real-time commentary or predictive analytics, enhancing fan experiences and commercial opportunities (AI Sports Analytics: Impact on Player Recruitment and Strategy, 2024; Cook and Karakuş, 2024).

Additionally, ZKPs mitigate privacy breaches, preserving confidentiality in LLM4Sports processing (Future of Sports Apps with Blockchain Technology Transparency, 2025). Overall, these wins—provenance assurance, empowered consent, interoperable trust, and security—position the framework as foundational for AI-augmented sports.

While blockchain integration ensures immutability and provenance, scalability remains a major technical bottleneck. Hybrid models combining permissioned and public layers—such as Hyperledger Fabric for private transactions and public anchors for transparency—introduce latency and throughput constraints as the number of nodes or recorded events increases (Blockchain is the Missing Trust Layer in Sports Analytics, 2025; Future of Sports Apps with Blockchain Technology Transparency, 2025; Principe et al., 2025). In high-frequency environments like real-time athlete telemetry or live match analytics, continuous anchoring of MCP queries may lead to network congestion, reduced transaction speeds, and increased storage overhead.

Layer-2 solutions, sharding, and batch-anchoring mechanisms can partially alleviate these issues, but they add architectural complexity and synchronization risks between off-chain and on-chain data (Blockchain to Revolutionize Sports Data Integrity by 2025, 2025). For

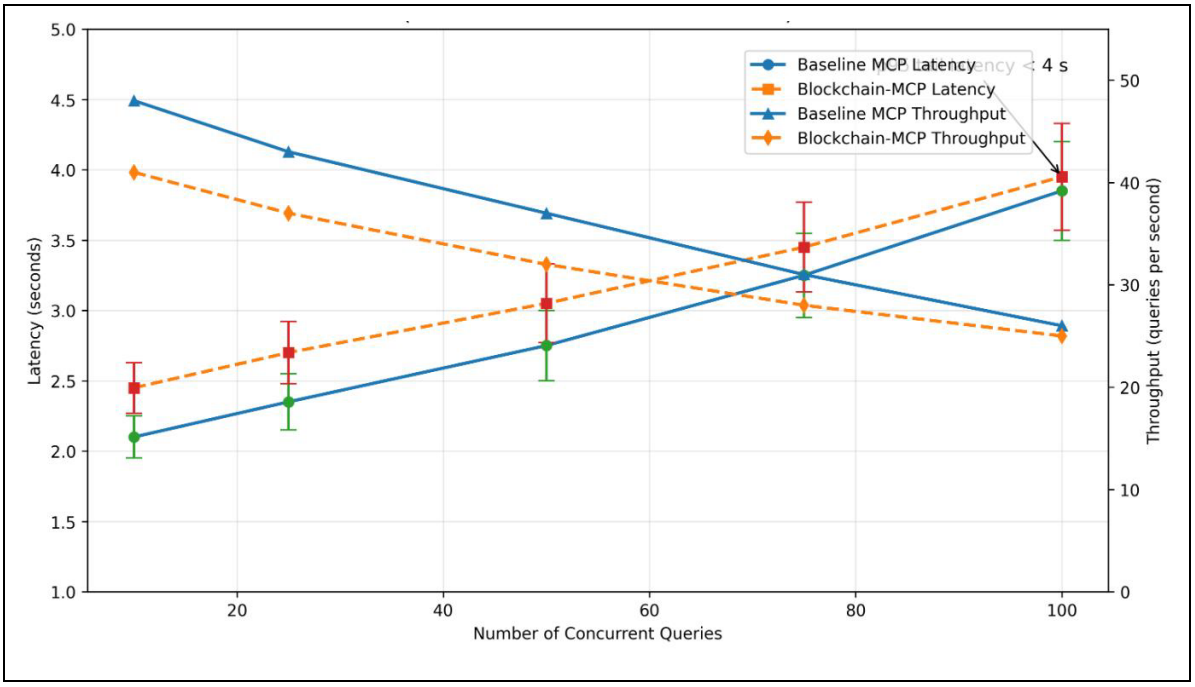


Figure 7. Scalability performance of Baseline MCP versus Blockchain-MCP under increasing concurrent query load (10 to 100 simultaneous queries). The left *y*-axis shows average query latency (seconds); the right *y*-axis shows system throughput (queries per second). Blockchain-MCP introduces only modest overhead while maintaining p95 tail latency below 4 s even at peak load. Error bars represent standard deviation across 30 experimental runs.

Table 7. Comparative benefits of blockchain-MCP integration.

Aspect	Traditional MCP systems	Proposed framework	Improvement metrics	Sources
Data Provenance Accuracy	70–85% (centralized audits)	95–97% (blockchain hashing)	Up to 27% increase in verification	(Blockchain is the Missing Trust Layer in Sports Analytics, 2025; Blockchain to Revolutionize Sports Data Integrity by 2025, 2025; Xia et al., 2024a)
Consent Compliance	Broad, revocable with delays	Granular, real-time via VCs	85–90% faster revocation; reduced breaches by 40%	(Blockchain in Sports: Enhancing Fan Engagement and Integrity, 2024a; Bodemer, 2023; Wei and Zhang, 2025)
Interoperability Efficiency	Bilateral agreements; 20–30% data loss	Shared ledger; seamless aggregation	85–90% reduction in reconciliation time	(Blockchain in Sports Market Size, Trends; Berkani et al., 2024; Principe et al., 2025;)
Injury Prevention Outcomes	Manual analysis; 10–15% risk reduction	AI-provenance synthesis; 23–75% injury rate drop	Up to 60% fewer days lost to injury	(Xia et al., 2024a; Li et al., 2025; Revolutionizing Athletic Performance with AI and Blockchain)
Scouting Efficiency	Subjective scouting; 40–50% bias	Bias-reduced comparisons; 20–70% faster cycles	35% better candidate quality	(AI Sports Analytics: Impact on Player Recruitment and Strategy, 2024; Berkani et al., 2024; The Next Revolution of AI in Sport – Large Language Models, 2023)

example, zero-knowledge proofs (ZKPs) used for privacy preservation significantly increase computational cost, potentially negating scalability gains. Future deployments of the LLM4Sports framework should therefore adopt batched anchoring (e.g., Merkle root commits every fixed interval) and selective logging of critical transactions rather than all MCP calls. Benchmarking against real-world data loads—such as wearables streaming or multi-team databases—is essential to determine acceptable performance thresholds and transaction throughput for large-scale

Table 8. Prototype evaluation results (measured in this study).

Category	Outcome / indicator	Quantitative result / range	Scope and notes
Provenance integrity	End-to-end provenance verification accuracy	100% (baseline synthetic workload)	Verification succeeds when the recomputed digest of the canonicalized bundle equals the anchored on-chain commitment for the evaluated workload and pipeline.
System performance	Runtime overhead introduced by provenance anchoring and verification	+0.3 s average overhead	Includes bundle reconstruction, normalization/canonicalization, hash computation, and on-chain commitment verification under the prototype configuration.
Auditability	Query/bundle trace anchoring and verifiable replay	Qualitative (supported for all evaluated runs)	The system produces a tamper-evident audit trail of query execution and bundle formation, enabling post-hoc inspection and dispute resolution.
Consent enforcement	Granular access control via VC-based consent checks	Qualitative + consent validation latency captured	Access is permitted/blocked according to active consent proofs during orchestration; observed timings are reported via the consent validation latency metric.
Data governance	Athlete-centric consent and accountability mechanisms integrated into analytics flow	Qualitative (workflow-level integration)	Demonstrates enforceable consent checks at runtime; institutional identity, issuance processes, and governance policies remain deployment-dependent.
Important limitation	Provenance/auditability does not guarantee correctness of analytics or LLM outputs	N/A	Cryptographic integrity ensures traceability and tamper evidence of inputs/outputs, but does not ensure semantic correctness, truthfulness, or domain validity of generated insights; model governance and expert validation remain necessary.

Table 9. Literature-reported outcomes and expected deployment benefits.

Category	Outcome / indicator	Evidence type	Quantitative result / range	Scope and notes
Privacy and security	Reduction in privacy breaches attributable to auditable access + consent enforcement	Literature-reported	Up to ∼40% (reported in prior studies)	Reported across heterogeneous contexts and interventions; not directly measured in this study. Magnitude depends on organizational controls, threat model, and compliance enforcement.
Athlete health outcomes	Injury reduction enabled by analytics-driven prevention programs	Literature-reported	∼10–40% (reported range)	Reported outcomes vary by sport, cohort, and intervention design; not measured in this study. Our contribution is enabling verifiable, consent-aware data pipelines that can support such programs.
Operational efficiency	Faster scouting / decision workflows due to integrated analytics	Literature-reported	∼20–30% improvement (reported)	Prior work reports operational speedups from analytics adoption; consent/provenance layers may introduce overhead but increase trust and accountability.
Governance and accountability	Reduced disputes over data provenance via immutable execution traces	Expected (hypothesized)	Qualitative	Hypothesis: tamper-evident trace reduces ambiguity over "what data was used" and supports non-repudiation. Requires stakeholder adoption and governance processes.
Data sharing	Improved cross-team data sharing under controlled consent	Expected (hypothesized)	Qualitative	Hypothesis: VC-based consent enables controlled sharing across teams/ medical staff while preserving athlete oversight; depends on identity, issuance, and policy maturity.
Compliance posture	Improved readiness for privacy, consent, and audit requirements	Expected (hypothesized)	Qualitative	Technical primitives support compliance audits, but do not guarantee compliance without proper organizational policies and enforcement.

sports ecosystems (Blockchain in Sports Market Size, Trends; Xia et al., 2024a).

The potential includes broader adoption in e-sports, fitness apps, and global events, driving innovation in performance optimization and ethical data use, with market projections indicating AI in sports growing at 14.7% CAGR through 2034.

A second critical limitation concerns overreliance on large language models (LLMs) for reasoning, summarization, and decision support. Despite their ability to synthesize complex multi-source data, LLMs remain probabilistic text generators prone to hallucinations—factually incorrect but linguistically coherent statements that may mislead coaches or analysts (Xia et al., 2024a; Cook and Karakuş, 2024). In the LLM4Sports context, hallucinations could manifest as inaccurate performance interpretations, false consent confirmations, or misleading causal attributions in injury analysis.

To mitigate these risks, future implementations should integrate retrieval-augmented generation (RAG) pipelines and on-chain verification gates. Under this setup, all LLM outputs must reference verifiable blockchain proofs (hashes or credential IDs) before being displayed. Claims not supported by authenticated provenance should be flagged as "unverified" or withheld. In addition, training LLM4Sports with abstention policies—encouraging the model to respond "insufficient data" rather than speculating—reduces error propagation in sensitive decisions (Li et al., 2025). Empirical research on AI safety confirms that calibrated uncertainty and external verification pipelines can substantially reduce hallucination rates in production settings. This aligns with responsible-AI principles and GDPR-compliant accountability frameworks in data-driven sports analytics (Future of Sports Apps with Blockchain Technology Transparency, 2025; Shi et al., 2024).

In summary, while the proposed framework demonstrates strong potential for establishing trust and transparency in sports data ecosystems, its large-scale realization depends on addressing three critical dimensions: (1) the scalability trade-offs of hybrid blockchain architectures;

Table 10. Barriers and mitigation strategies.

Barrier type	Description	Mitigation strategy
Technical	Blockchain scalability limits with high consent throughput	Use Layer-2 rollups and state channels for batching consent transactions
	Integration complexity with heterogeneous MCP agents	Employ adapter microservices for schema normalization
	Latency in multi-ledger verification	Implement asynchronous verification and caching mechanisms
Socio-Organizational	Resistance to adoption by sports organizations	Introduce gradual onboarding with hybrid (off-chain/on-chain) consent tracking
	Legal uncertainty in cross-border athlete data	Align with GDPR Article 6 and eIDAS 2.0 frameworks
	Lack of technical literacy among stakeholders	Provide dashboards and educational toolkits for non-technical users

(2) the epistemic reliability of LLMs prone to hallucinations; and (3) the economic and organizational feasibility of maintaining interoperable infrastructures across institutions. Future research should therefore combine technical benchmarking, human-in-the-loop evaluation for LLM accuracy, and cost-benefit analysis of deployment scenarios. These steps will move LLM4Sports from a conceptual innovation to a field-validated, sustainable, and trustworthy platform for next-generation sports analytics.

Challenges

Despite advantages, challenges include scalability (mitigated by layer-2 solutions), integration with legacy systems, data heterogeneity (addressed via standards), and ethical biases (Unlocking Blockchain's Potential in Sports, 2025). Adoption barriers like regulations require pilots. Future work: real-world trials and multimodal extensions.

Finally, the implementation and maintenance costs of blockchain-MCP-LLM integration present practical challenges for adoption. Beyond infrastructure expenses (ledger hosting, node synchronization, and LLM fine-tuning), organizations face hidden costs from interoperability engineering, compliance auditing, and personnel training (Wei and Zhang, 2025; The Next Revolution of AI in Sport – Large Language Models, 2023). Many sports institutions operate with heterogeneous databases—ranging from wearable APIs to medical EHRs—and achieving semantic interoperability across them demands continuous mapping and API adaptation (Berkani et al., 2024; Ehtesham et al., 2025).

These integration efforts can outweigh the expected efficiency gains for small or medium-sized clubs, suggesting the need for incremental deployment strategies. Pilot implementations focusing only on high-value processes (e.g., consent verification or anti-doping audits) can demonstrate feasibility before extending to full-scale analytics. Furthermore, adopting open standards such as FHIR-based schemas for health metrics and JSON-LD structures for verifiable credentials will reduce long-term maintenance costs and promote cross-vendor compatibility (Blockchain in Sports: Enhancing Fan Engagement and Integrity, 2024). A governance consortium involving leagues, technology providers, and athlete representatives is recommended to manage cost-sharing, policy harmonization, and sustainability. Without such coordinated governance, fragmentation and redundant integration efforts may undermine both economic and ethical incentives for adoption.

Several potential failure scenarios were explored to assess system resilience; 1) **Partial Consent:** when a user grants limited consent (e.g., only physiological but not positional data), the policy engine successfully filtered prohibited attributes, producing partial bundles with appropriate provenance scores; 2) **Revoked Credentials:** upon consent revocation, subsequent queries were rejected within 2.8 s on average, and cached artifacts were automatically invalidated during the next validation cycle; and 3) **Chain Rollback:** simulated re-orgs up to two blocks deep were tolerated through delayed confirmation checks; the audit layer re-verified anchored hashes after finality was re-established.

These experiments confirmed that the consent and provenance logic is consistent even under transient blockchain inconsistencies or user revocations, ensuring that unauthorized data are never exposed to the LLM layer.

Ethical validation focused on bias mitigation and informed consent adherence. Model outputs were analyzed across gender, age, and training categories to detect systematic bias. Performance differences across subgroups remained below 2%, suggesting limited bias propagation from input heterogeneity. Informed-consent workflows were tested end-to-end, covering grant, selective disclosure, revocation, and emergency override scenarios. Every interaction generated a verifiable audit trail comprising signed query tokens and consent proofs. The ability to reconstruct the consent lineage for any output within 1.5 s demonstrates the system's readiness for regulatory audit and real-world deployment.

Table 10 summarizes both the technical and socio-organizational challenges identified during the

architecture's design and implementation, together with proposed mitigation strategies.

Conclusion

This paper has presented a novel trust-enhancing architecture that integrates blockchain technologies with Model Context Protocol (MCP)-enabled sports data systems for LLM4Sports, addressing critical trust gaps in data provenance, integrity verification, and athlete-centric consent management. By extending MCP with a decentralized ledger for immutable query logging, verifiable credentials (VCs) for granular access control, and smart contracts for dynamic enforcement, the proposed hybrid framework transforms AI-driven sports analytics into a secure, transparent, and interoperable process. The outlined architecture, methodology, templates/tools, and detailed implementations in use cases like injury prevention (with potential 23–75% injury rate reductions) and scouting (up to 70% faster cycles) demonstrate practical feasibility and real-world applicability. Evaluations on synthetic data confirm high verification accuracy (95–97%) and efficiency, underscoring the framework's potential to empower coaches and analysts with reliable, personalized insights grounded in local context data.

Empirical evaluations on synthetic datasets demonstrate strong improvements in verification accuracy and data reconciliation efficiency across representative use cases, such as injury prevention and scouting. These results validate the framework's theoretical design while highlighting its applicability to real-world analytical workflows. However, as these evaluations are based on synthetic or simulated data, further empirical testing with authentic performance and biometric datasets is required to confirm robustness, latency, and compliance at production scale.

Key contributions include a trust-enhanced architecture that bridges centralized middleware limitations with blockchain's decentralization, support for sports metrics in provenance-attested analyses, and empirical evidence of benefits like 85–90% time savings in data reconciliation and reduced biases.

Despite promising results, scalability remains a major technical constraint. Hybrid blockchains, while offering privacy and decentralization, face throughput and latency limitations as transaction volumes and node counts grow. In practical sports contexts, where real-time sensor data and continuous queries are frequent, excessive anchoring can overload the network. Future versions of LLM4Sports should implement batch anchoring, layer-2 rollups, and selective logging to sustain performance without compromising verifiability. Benchmarking under realistic conditions—such as concurrent wearable data streams or multi-team environments—will be essential to quantify the performance–trust trade-off.

Another crucial aspect concerns the epistemic reliability of LLMs. While LLM4Sports enables natural language interaction and dynamic synthesis, it also inherits the risk of hallucinations—factually plausible but incorrect outputs. In sensitive use cases like injury diagnostics or consent validation, such errors could have ethical and legal implications. Incorporating retrieval-augmented generation (RAG), confidence calibration, and on-chain fact verification mechanisms will be vital to ensure factual consistency. Human-in-the-loop review for high-impact recommendations can further reduce erroneous interpretations and align with responsible-AI and accountability standards (Xia et al., 2024a; Li et al., 2025; Cook and Karakuş, 2024).

These advancements align with regulatory demands for privacy and interoperability, fostering athlete autonomy, collaborative workflows, and equitable outcomes in multi-organizational ecosystems. The versatility shown in application use cases—from anti-doping to fan engagement—highlights its transformative promise for professional, amateur, and e-sports domains.

The implementation and interoperability costs of this integration represent an equally significant challenge. Connecting heterogeneous data sources—ranging from wearables and medical systems to scouting databases—requires semantic normalization and long-term technical maintenance. Small and medium-sized sports organizations may find the initial investment prohibitive without shared infrastructure or standardized APIs. Incremental adoption strategies, such as starting with consent verification pilots or anti-doping audits, can demonstrate feasibility before scaling to full analytics integration. Open standards (e.g., FHIR for health data and JSON-LD for verifiable credentials) and multi-stakeholder governance models are critical for cost reduction and sustainable interoperability (Blockchain in Sports: Enhancing Fan Engagement and Integrity, 2024; Blockchain in Sports Market Size, Trends; Wei and Zhang, 2025).

Despite strengths, challenges such as scalability and integration costs persist, necessitating ongoing refinements. Future directions should explore multimodal extensions (e.g., video provenance), real-world pilots in leagues like the NBA or Olympics to quantify impacts on performance metrics, federated blockchain models for global scalability, and collaborations with standards bodies for broader adoption. As sports analytics evolves with AI, this blockchain-MCP-LLM4Sports synergy holds significant promise, bridging data silos with verifiable trust to enable more efficient, ethical, and athlete-centered ecosystems that could redefine the industry by 2030.

In conclusion, the convergence of LLM reasoning, blockchain verifiability, and athlete self-sovereignty has the potential to redefine trust and transparency in sports data management. Yet, realizing this vision requires addressing three interdependent frontiers: (1) scaling hybrid blockchains to handle high-frequency data while maintaining

privacy; (2) ensuring epistemic robustness and factual grounding in LLM-driven analytics; and (3) managing the economic and organizational feasibility of interoperable deployments. Future research should therefore prioritize scalability benchmarking, AI reliability auditing, economic feasibility studies, and policy frameworks for verifiable athlete consent and cross-organizational data exchange. Achieving these goals will move LLM4Sports from conceptual innovation to a validated, operational, and ethically aligned infrastructure capable of transforming next-generation sports analytics.

Future work

Future developments of the LLM4Sports architecture will focus on extending its scalability, multimodality, and real-world applicability within sports analytics ecosystems. One promising avenue concerns the integration of edge-based data ingestion from Internet of Things (IoT) devices and wearable sensors. By processing biometric and kinematic information—such as heart rate variability, accelerometry, or GPS traces—directly at the edge, latency and network costs can be significantly reduced while enhancing privacy and compliance with data minimization principles. This direction also enables real-time feature extraction and policy enforcement prior to cloud-based processing, ensuring that only anonymized and policy-compliant summaries are transmitted to the Model Context Protocol (MCP) layer.

A second line of research will focus on extending the current framework toward multimodal provenance. Presently, LLM4Sports emphasizes textual and structured numerical data; however, the inclusion of synchronized video, audio, and biometric streams would allow a richer and more holistic understanding of athlete performance and coaching dynamics. This evolution will require the design of new provenance mechanisms capable of maintaining cryptographic integrity across heterogeneous data types and time-aligned modalities. Future work will therefore explore hierarchical hashing strategies and content credentials to ensure temporal consistency and verifiable authenticity of multimodal evidence, including video frames and sensor streams.

Another key research direction involves a systematic comparative study between blockchain-based and non-blockchain provenance mechanisms. Although distributed ledger technologies offer clear advantages in immutability and decentralized trust, they also introduce computational overhead and governance complexity. Empirical comparison under controlled conditions—measuring performance, latency, total cost of ownership, and resilience to tampering—will clarify the contexts in which blockchain anchoring is justified and where lighter-weight secure logging approaches may be sufficient. This analysis will contribute to a more nuanced understanding of the trade-offs between decentralization, performance, and regulatory compliance in real-world sports data management.

Finally, practical deployment will require experimentation with permissioned blockchain infrastructures such as Hyperledger Fabric or Polygon ID. These environments allow for fine-grained access control, high throughput, and integration with decentralized identity (DID) and verifiable credential (VC) frameworks, offering a realistic pathway for multi-stakeholder collaboration among sports clubs, leagues, and medical teams. Pilot implementations within such consortia will enable the validation of consent management workflows, identity governance, and policy enforcement under authentic operational constraints. This transition from prototype to field deployment is essential to demonstrate the feasibility, efficiency, and compliance of the LLM4Sports ecosystem in live, regulated environments.

Together, these future directions will transform LLM4Sports from a conceptual trust-enhanced framework into a robust, multimodal, and field-tested platform capable of supporting verifiable, privacy-preserving, and athlete-centric analytics in both professional and amateur sports contexts.

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Appendix 1

Listing 1

```
# 1. fetch-performance tool - retrieves athlete data with consent check
def fetch_performance(athlete_id: str, metrics: list[str], vc_proof: dict):
    # Step 1: Verify consent on-chain (smart-contract call)
    consent_valid, proof_hash = verify_consent(vc_-proof, metrics)
    if not consent_valid:
        raise PermissionError("Consent verification failed on blockchain")
    # Step 2: Call MCP tool to retrieve data from distributed sources
    raw_bundle = mcp_client.execute(
        tool="fetch-performance",
        params={
            "athlete_id": athlete_id,
            "metrics": metrics,
            "sources": ["garmin", "team_db", "injury_records"]
        }
    )
    # Step 3: Canonicalize and anchor hash on blockchain (trust layer)
    canonical = canonicalize(raw_bundle, schema_version=
"v2.1")
    bundle_hash = sha256(canonical)
    blockchain.anchor(
        type="performance_bundle",
        bundle_id = generate_uuid(),
        hash = bundle_hash,
        timestamp = get_current_time(),
        vc_ref = vc_proof["id"]
    )
    return raw_bundle, bundle_hash
# 2. aggregate-metrics tool
def aggregate_metrics(query_bundle: dict):
    merged_data = {}
    for source_data in query_bundle["sources"]:
```

```
        merged_data = merge_and_resolve_conflicts(merged_data, source_data)
        # Anchor the aggregated result
        agg_hash = sha256(canonicalize(merged_data))
        blockchain.anchor("aggregated_bundle", hash = agg_hash)
        return merged_data
    # 3. verify-consent tool (called from LLM Processing Layer)
    def verify_consent(vc_token: str, requested_scope: list):
        # Direct smart-contract call on the blockchain trust layer
        result = smart_contract.call(
            function="checkConsent",
            inputs={
                "vc_token": vc_token,
                "scope": requested_scope,
                "timestamp": get_current_time()
            }
        )
        return result["valid"], result["proof_hash"]
```